

Advanced Kinematics

Body parts used

- **Jerk (linear):** all body parts (computed from each part's velocity/acceleration, then aggregated).
 - **Angular acceleration:** orientation from **bodypart2** → **bodypart16** (head → tail).
 - **Frequency-domain features:** whole-fish signals derived from **all body parts** (e.g., Bp_velocity_mean, Summed_movement).
 - **Rhythmic patterns:** same whole-fish signals (autocorr / peak timing).
 - **Complexity metrics:** whole-fish signals (symbolized headings or speed series).
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1) Jerk & Angular Acceleration

Linear jerk (change in linear acceleration)

it measures: how quickly linear acceleration itself is changing.

- **{bp}_jerk_{W}_window**
 Δ of {bp}_acceleration_{W}_window compared to W frames earlier.
Units: $\Delta(\text{mm in last 1 s})$ across W frames.
 - **Bp_jerk_mean_{W}_window, Bp_jerk_stdev_{W}_window**
Mean / stdev across all body-part jerks for that W.
Units: $\Delta(\text{mm in last 1 s})$ across W frames.
 - **Body parts used:** all body parts (one per-part jerk, then aggregated).
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Angular acceleration (how fast turning rate changes)

What it measures: change in heading rate (second derivative of angle).

- **Angular_velocity**
Framewise wrapped angle change $\times$ fps, in **deg/s** (derived from bodypart2 → bodypart16).
- **Angular_acceleration_{W}_window**
 Δ of Angular_velocity vs W frames earlier.
Units: **deg/s²** (because the Δ in deg/s over W frames is normalized by time).

- **Angular_acceleration_abs_{W}_window**
Absolute value (magnitude) of angular acceleration over the same window.
Units: deg/s².
- **Bp_angular_acceleration_mean_{W}_window,**
Bp_angular_acceleration_stdev_{W}_window
Whole-trace summaries of Angular_acceleration_{W}_window.
Units: deg/s².

Body parts used: bodypart2 & bodypart16 (to form orientation).

2) Frequency-Domain Features

Signals analyzed (per window W):

- **Bp_velocity_mean** (average per-part velocity; $\approx$ mm/s).
- **Summed_movement** (sum of per-part movement; mm/frame).

We compute a PSD (e.g., Welch) over each sliding window W, then report the dominant rhythms.

For each **signal** SIG $\in$ {Velocity, Movement} you'll see:

- **{SIG}_dominant_freq_{W}**
Highest-power frequency in the window.
Units: Hz.
- **{SIG}_dominant_freq_power_{W}**
PSD power at that frequency.
Units: PSD power (arbitrary, but comparable within your setup).
- **{SIG}_secondary_freq_{W}, {SIG}_secondary_freq_power_{W}**
Second-strongest frequency and its power.
Units: Hz, PSD power.
- **{SIG}_spectral_entropy_{W}**
Entropy of the normalized spectrum (how “broadband” it is).
Units: bits (log base 2).

Body parts used: indirectly **all**, via Bp_velocity_mean and Summed_movement.

Examples:

Velocity_dominant_freq_150, Velocity_dominant_freq_power_150,
Movement_secondary_freq_150, Movement_spectral_entropy_150

3) Rhythmic Patterns (time-domain periodicity)

What it measures: timing and strength of repeating patterns in speed/movement using autocorrelation and peak detection (per window W).

For each **signal** SIG $\in$ {Velocity, Movement}:

- **{SIG}_rhythm_period_{W}**
Lag (in **seconds**) of the first prominent autocorr peak (dominant cycle length).
Example: 0.33 s $\approx$ 3 Hz rhythm.
- **{SIG}_rhythm_strength_{W}**
Autocorr value at that peak (0–1). Higher = cleaner rhythm.
- **{SIG}_inter_peak_interval_mean_{W}, {SIG}_inter_peak_interval_std_{W}**
Mean / std of intervals between detected peaks (in **seconds**).
- **{SIG}_rhythm_CV_{W}**
Coefficient of variation of inter-peak intervals (std/mean).
Units: dimensionless (lower = steadier pacing).

Body parts used: indirectly **all**, via the whole-fish signals.

Examples:

Velocity_rhythm_period_150, Velocity_rhythm_strength_150,
Movement_inter_peak_interval_mean_150, Movement_rhythm_CV_150

4) Complexity Metrics

What it measures: how “structured” vs “random” the motion/heading is (per window W).

- **Velocity_sample_entropy_{W}**
Sample entropy of Bp_velocity_mean in the window.
Units: dimensionless (higher = more irregular).
- **Velocity_permutation_entropy_{W}**
Permutation entropy of Bp_velocity_mean.
Units: dimensionless.

- **Heading_LZ_complexity_{W}**

Lempel–Ziv complexity of the **compass-digit** sequence (0–7) over the window.

Units: dimensionless (higher = more pattern novelty).

Body parts used:

- Velocity entropies: **all** (through Bp_velocity_mean).
- Heading complexity: **bodypart2 & bodypart16** (orientation → compass digits).

Examples:

Velocity_sample_entropy_150, Velocity_permutation_entropy_150,

Heading_LZ_complexity_150
